

Title

Spring 2026 Data Science Project

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Contributions:

- Nina Khilnani
- Diya Peechara
- Rahil Shiraz
- Maanya Upadhyay
- Saachi Samdani

For each member, list which of the following sections they worked on, and summarize the contributions in 1-2 sentences. Be specific!

A: Project idea

B: Dataset Curation and Preprocessing

C: Data Exploration and Summary Statistics

D: ML Algorithm Design/Development

E: ML Algorithm Training and Test Data Analysis

F: Visualization, Result Analysis, Conclusion

G: Final Tutorial Report Creation

H: Additional (not listed above)

Introduction:

The purpose of this project is to analyze felony sentencing in the District of Columbia using the entire data science pipeline. A main goal of the United States criminal justice system is to ensure equal protection under the law for all individuals. However, many are concerned that bias is prevalent throughout the steps of the criminal justice system, especially sentencing. We are interested in understanding whether factors such as race, gender, age, etc. are significant contributors to sentencing outcomes within D.C. Our overall topic first addresses the key attributes that influence sentencing lengths in general. While we expect attributes such as the type of offense committed or the severity of the crime to be the best predictors, we are also interested in whether or not demographic attributes have any impact on sentencing lengths or even offense types. We are also curious about general trends regarding crime, whether that be the relationship between certain demographics and qualities of a crime, or even types of crimes and their relationship with sentencing. Answering these questions are important in better understanding bias in criminal justice systems. Identifying such issues using a data driven methodology is the first step towards solving such issues.

Data Curation and Preprocessing

The dataset we primarily use throughout the course of this project is the “Felony Sentences” dataset from Open Data DC. We selected this dataset due to our interest in learning more about the criminal justice system in the D.C. area. This dataset is also large enough for us to analyze a number of different variables, regarding demographics, different classes of offenses, as well as the resulting sentences and outcomes. The dataset is downloaded in csv format and the Python language is used for the analysis of this data. After loading the data sets, we applied a general `clean_dataset()` function to all 3 files, which gets rid of whitespace, converts to lowercase, and replaces spaces with underscores, removes duplicate rows, and drops rows where key identifiers are missing. Columns that have dates are parsed using `pd.to_datetime()`, and string columns are converted to numeric types when possible. Then, the column median replaces missing values for numeric columns and “Unknown” is put in for the non-numerical ones. Then, we standardized the main fields to all uppercase strings to remain consistent. However, the dataset included many columns that were not informative, such as `rid`, `generic_case_id`, `generic_offender_id`, `gis_id`, and a few other id columns. Therefore, we dropped those columns and instead created 2 new columns: `total_fine`, the sum of `vvca_amt`, `fine_amt`, and `restitution_amt` to show the total financial penalty of a sentence, and `serve_ratio`, which is the ratio of `sentence_to_serve_months` to `sentence_imposed_months`, representing the proportion of the sentence that a defendant is expected to serve. After completing the preprocessing, we verify that the final dataset contains no null values through `df.isna().sum()`.

Exploratory Data Analysis

Before applying the machine learning models, we must explore the dataset through hypothesis testing and visualization to understand how the variables are related to each other. This section gives us a baseline understanding of the data and also helps validate our hypothesis. We focus specifically on relationships between demographic variables and sentencing outcomes, as this is what our research questions test.

Hypothesis Testing

We utilize three different statistical methods for initial hypothesis testing. For the first method, we use a chi squared test in order to assess the relationship between gender and offense type. The null hypothesis for this test is that the gender of the individual does not have an effect on the type of offense they commit, with the alternative hypothesis being that gender does have an effect on the type of offense. A chi squared test is used because both variables are categorical. The low resulting p-value allows us to reject the null hypothesis that the gender of the individual does not have an effect on the type of offense they commit. To supplement this test, we create a grouped bar graph that displays the number of each type of offense committed (in the dataset) among each gender. The plot reveals that a large proportion of offenses in the dataset are committed by male individuals.

For the second hypothesis test, we use the t-test method. The null hypothesis is that the gender of the individual does not have an effect on the length of sentencing, with the alternative

hypothesis being that the gender of the individual does have an effect on the length of sentencing. Based on the low p-value, we reject the null hypothesis. Paired with this test is another grouped bar graph displaying different offense types and their average sentence length, grouped by gender. This plot is interesting because it reveals that even among the same offense types, there is a difference in sentence length between male and females. For instance, the average sentence length for a male individual who committed homicide is over 200 months, while the average sentence length for a female who committed homicide is under 150 months. This implies a bias in sentencing length.

Our final statistical method used for hypothesis testing is an analysis of variance (ANOVA). The null hypothesis for this test is that the type of offense does not affect the average length of sentencing, while the alternative hypothesis is that type of offense does have an effect on the average length of sentencing. As expected, the low p-value and large F-statistic allows us to reject the null hypothesis. To accompany this test, we created boxplots for each type of offense to display the distribution of sentence lengths. Here we see that homicides have significantly higher average sentence lengths than any other crime, due to the severity of the crime.

Overall, these tests provide us with some baseline understanding of our data. It allows us to further understand important relationships between gender, offense types, and sentence lengths. This teaches us more about demographic trends in crime, as intended. It also serves as a sanity check, as we found that offense type influences sentencing lengths, as expected.

Primary Analysis and Visualizations

Our overarching goal for the machine learning section is to identify the key attributes that influence felony sentencing lengths in D.C., and to assess whether demographic factors like race, gender, and age play a role. To begin, we define our target and feature variables. The target variable is the sentence length in months ('sentence_imposed_months'), while our feature variables include charge number, year of sentence, race, gender, age, offense, offense type, homicide type, and offense severity group. We use one-hot encoding in order to handle the categorical features. Next, we split the data into a training and testing set in order to prevent overfitting.

For this section, we chose linear regression over classification or clustering. Linear regression seemed to be the most useful for a number of reasons. First, our target variable is continuous, the sentence length seems to range from 0 to 840 months. Linear regression naturally seemed better to predict the continuous variable outcomes. Secondly, our EDA showed that there was likely a linear relationship. The ANOVA test showed that the offense type had a strong effect on sentence length. For these reasons, linear regression seemed to be the best approach to predicting the variable.

The first model we employ is a linear regression to predict sentence length. We start by fitting the model on the training data and then predicting the target variable using the test set. In order to evaluate the model we calculate the mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and R-squared (R^2). The R^2 is 0.79, which means that

the model explains about 79% of the variance in sentence length, which is a strong result. The MAE, 11.63 months, tells us that on average, our predictions are off by a little less than a year. This result is not negligible, but it is reasonable because the sentence lengths in the data set are from 0 to 840 months and have a standard deviation of 53 months. The RMSE, 25.37, is higher than the MAE, meaning the model makes large errors occasionally.

We also chose to display the top 10 features by coefficients (absolute value) from the linear regression. These features are the ones that influence expected sentence length the most, as determined by the linear regression model. Essentially, holding all other features constant, if a crime/observation has a specific feature (value of 1 or True for that column), the predicted sentence length in months increases by the amount of its coefficient, relative to an observation not having that feature. As seen below, the top 10 features are all related to the type of offense of the crime. The accompanying visualization plots the coefficient values of the top 20 features in order to help the reader better understand even more features that have an impact on the target variable. This plot allows the reader to see that the most important features all have to do with the type of crime committed, as expected. We expand on these results with the subsequent models.

The second model we employ is random forest with 100 decision trees. After seeing the results, we learn that the random forest regressor performs almost identically to linear regression. It has a slightly better R^2 of 0.8 and a slightly lower RMSE of 25.14 months. The RMSE remains the same, which is 25.37 months. Due to the similarity and consistency of these 2 different models, our confidence in these results is strengthened. However, the random forest features are a little different than the linear regression coefficients. Homicide_type_unknown, 0.278, offense_severity_group_M3, 0.104, and charge_number, 0.083, rank the highest, and then offense_type_Homicide and other severity groups. The homicide related features and severity groups are most prominent, which shows us how random forest unlike linear regression shows us how features interact with each other. This demonstrates that homicide classification and severity together are influential features.

The third model we used is a 2-layer neural network over 50 epochs. Its performance metrics are an MAE of 11.35 months, RMSE of 24.31 months, and an R^2 factor of 0.81. Based on these metrics, the neural network has the best performance of the three models so far, with the lowest MAE and the highest R^2 . Even though there isn't much improvement from the linear regression model, this small improvement shows us that there is likely a linear relationship between our features and the sentence length and that the added complexity of the neural network only captures marginal additional signal. Also, the training and validation loss curve converge over epochs, showing us that the model is not overfitting. Across all three models, the R^2 ranging from 0.79-0.81 shows us that our feature set captures the majority of what influences sentence length.

The fourth model we used is a linear regression using a broader and reduced feature set. It follows the same methodology as the first model, but it excludes offense type, homicide type,

and offense severity group. This means it only has the columns charge number, sentence year, race, gender, age group, and the specific offense. This allowed us to isolate the demographic and charge-level features and see their effect without the influence of severity data. The results show that there is a drop in performance. The R^2 falls to 0.73, MAE increases to 14.10 months, and RMSE rises to 28.98 months. This confirms that the severity of the offense and type are critical factors in predicting sentence length. However, the fact that this reduced model still has an R^2 value of 0.73 is a significant finding. Demographic variables combine with the specific offense explain almost $\frac{3}{4}$ of the variance in sentence length. This supports our hypothesis that demographic factors do play a measurable role in sentencing outcomes.

Our fifth and final model predicts the serve ratio, which is the proportion of an imposed sentence that a defendant is actually expected to serve. This is a highly important distinction that the sentence length cannot really capture. We trained a linear regression model using the same complete features as Model 1, with the addition of sentence type. The results were more surprising: the R^2 is 0.97, MAE is 0.034, and RMSE is 0.074. This means our model predicts the serve ratio with amazing accuracy. This suggests that the proportion of a sentence actually served is determined by the factors like sentence type, offense severity, and offense classification. There is very little unexplained variance. This is different from the sentence length, because in that, the demographic variables contribute to predictions more meaningfully. Together, models 4 and 5 both show much more in relation to the predictions that demographic characteristics appear.

Insights and Conclusion

To answer our core research questions, we trained five models on the DC Felony Sentencing dataset. Below is a summary of our results:

Model	Target	MAE	RMSE	R^2
Linear Regression	Sentence Length (month)	11.63	25.37	0.79
Random Forest	Sentence Length (month)	11.64	25.14	0.80
Neural Network	Sentence Length (month)	11.35	24.31	0.81
Linear Regression (broader features)	Sentence Length (month)	14.10	28.98	0.73
Linear Regression	Serve ratio	0.034	0.074	0.97

Following are the key insights we found:

- (1) Offense type and severity are the dominant predictors of sentence length. This did not surprise us, as severity and type of offense naturally predict sentence length. Every model we trained reflected this, the features related to the severity of the crime were consistently

the most influential in predicting sentence length. This is what we initially expected and also shows that the legal system is working as intended for DC sentences.

- (2) Removing offense-level features significantly weakens predictions. Model 4, which we trained very similar to Model 1 except for the fact that it only included broader features, showed a drop in R^2 from 0.79 to 0.73, which means that demographic variables like race, gender, and age do influence the variance in sentence length.
- (3) All three types of models perform consistently. Linear regression, random forest, and the neural network achieved R^2 values of 0.79, 0.80, and 0.81 respectively, with RMSE values ranging between 24 and 25 months. The fact that these models performed almost identically proves that the relationships in the data are mostly linear.
- (4) Serve ratio is predictable. Model 5 had the highest performance by far. This means that given the features available, we can predict the proportion of a sentence the defendant will actually end up serving with very high accuracy.

Limitations

- Correlation is not causation. The fact that demographic variables seem to change the models does not mean that the judges are making decisions based on race or gender. There may be confounding variables that we have not analyzed that end up changing our results.
- This dataset only covers the DC area. Our findings are specific to this area so we cannot generalize them to other jurisdictions which have different guidelines or sentencing.
- Missing data changes our information. The missing values were filled with column medians and categorical missing values were filled with "Unknown." This may change the model performance, especially for columns where there is a lot of missing data (like homicide_type).
- Arrests and crimes datasets were not used in final models. While we loaded and cleaned these datasets with the intention of merging them with the sentencing datasets, time constraints limited our ability to merge them with sentencing data. Incorporating these datasets in the future can tell us much more about the context of the crimes and how sentences ended up.

Conclusion

At the start of this project, we wanted to answer two major questions:

- (1) what factors most strongly influence felony sentencing outcomes in DC? And
- (2) do demographic characteristics play a role in sentencing outcomes?

Our project clearly answers both of these questions.

Offense type and severity are the strongest predictors of sentence length, but they do not account for the entire outcome. Our hypothesis tests showed us that there were statistically significant differences in sentencing outcomes across gender groups even when the sentences had the same

offense categories. While we cannot conclude the reason for this bias, the patterns that we found are consistent with broader concerns about fairness in the criminal justice system and if we had more time, this is something we would research further.

The most surprising outcome of our project is how predictable the serve ratio is. The R^2 value of 0.97 suggests that once the sentence is handed, how much of it the defendant actually serves is determined by rule-based factors. By contrast, the sentence length itself has some variance that is explained by demographic factors.

We believe that analyzing this issue with data is a good starting point for actually understanding and investigating these issues in the legal system. By making our methodology transparent, we hope that this is reproducible and others can further our research about the power and responsibility that can be made clear by data science.

Project Instructions : For your final semester long group project, you will publish a tutorial that will walk users through the entire data science pipeline: acquiring and curating data, parsing the data into a queryable format, exploratory data analysis, hypothesis testing and machine learning, and explaining the results with words and visualizations. This project is designed to demonstrate your data science knowledge and contribute to your portfolio

Requirements:

- (15 points) Formatting and prose.
 - Follow the format given at the end of this section. Use section headers.
 - For each section, include a clear explanation of what you are doing. We will be checking the entire tutorial, including the parts you submitted in previous checkpoints.
 - Write code that is documented, well-organized, and reproducible.
 - Does it help the reader understand the tutorial?
 - Cite your sources: Link to other resources that would: (a) give a lagging reader additional help on specific topics, and (b) give an advanced reader the opportunity to dive more deeply into a technique or idea.
- (30 points) Machine learning analysis that will help you answer the questions you posed in the introduction.
- (10 points) Visualization based on your results.
- (15 points) Insights and conclusions.

1. Header with contributions.

Title

Spring 2026 Data Science Project

Your name(s)

Contributions: For each member, list which of the following sections they worked on, and summarize the contributions in 1-2 sentences. Be specific!

A: Project idea

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C: Data Exploration and Summary Statistics

D: ML Algorithm Design/Development

E: ML Algorithm Training and Test Data Analysis

F: Visualization, Result Analysis, Conclusion

G: Final Tutorial Report Creation

H: Additional (not listed above)

2. Introduction. The introduction should motivate your work: what is your topic? What question(s) are you trying to answer with your analysis? Why is answering those questions important?
 - a. Topic
 - i. Felony sentencing in D.C
 - ii.
 - b. Questions we want to answer
 - i. What influences the type and length of a sentence of a crime in D.C.
 1. Manual check of the data → see that many of the same types of crime with the same severity ratings face different sentences
 - a. Curious on why that may be
 - c. Why is it important
 - i. Better understand if there is demographic bias in the criminal justice system in D.C (and how extreme it may be)
 1. Racial, gender, age...
3. Data curation. Cite the source(s) of your data. Explain what it is. Transform the data so that it is ready for analysis. For example, set up a database and use SQL to query for data, or organize a pandas DataFrame.
 - a. Checkpoint 1 and 2
4. Exploratory data analysis. (See checkpoint 2.)
 - a. Might have to add to to connect to next part
5. Primary analysis. Based on the results of your exploration, choose a machine learning technique (e.g., classification, regression, clustering, etc.) that will help you answer the questions you posed in the introduction. Explain your reasoning.
 - a. Ideas
 - i. Classification
 1. Sentence type: incarceration or not?

- ii. Regression
 - 1. Predicted length of sentence?
 - 2. Then compare across racial groups, gender, etc.
 - 3. What feature has the biggest impact on sentence length?
 - a. Multiple regressions.
 - b. Can probably use the sentences dataset for this and then supplement with the arrest/crime in 2016 if needed to explain our findings
- 6. Visualization. Explain the results and insights of your primary analysis with at least one plot. Make sure that every element of the plots are labeled and explained (don't forget to include a legend!).
 - a.
- 7. Insights and Conclusions. After reading through the project, does an uninformed reader feel informed about the topic? Would a reader who already knew about the topic feel like they learned more about it?